

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{9} g_2^4 Y_e^{i1i2} \text{LF}_{3,0}[\mathbf{m}_2] - \frac{1}{6} g_2^4 Y_e^{i1i2} \text{LF}_{4,-1}[\mathbf{m}_2] + \frac{8}{45} g_2^4 Y_e^{i1i2} \text{LF}_{5,-2}[\mathbf{m}_2] + \frac{1}{72} g_2^2 Y_e^{i1i2} \right. \\
& \quad \left(18 \overline{Y}_e^{\text{pr}} Y_e^{\text{pr}} - 13 \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{3,0}[\mathbf{m}_1^{\text{p}}] + \frac{1}{48} g_2^2 Y_e^{i1i2} \left(-12 \overline{Y}_e^{\text{pr}} Y_e^{\text{pr}} + 11 \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{4,-1}[\mathbf{m}_1^{\text{p}}] - \\
& \quad \frac{2}{45} \sum_{\mathbf{p}} g_2^4 Y_e^{i1i2} \text{LF}_{5,-2}[\mathbf{m}_1^{\text{p}}] + \frac{1}{24} g_2^2 Y_e^{i1i2} \left(18 \overline{Y}_d^{\text{pr}} Y_d^{\text{pr}} + 18 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} - 13 \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{3,0}[\mathbf{m}_q^{\text{p}}] + \\
& \quad \frac{1}{16} g_2^2 Y_e^{i1i2} \left(-12 (\overline{Y}_d^{\text{pr}} Y_d^{\text{pr}} + \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}}) + 11 \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{4,-1}[\mathbf{m}_q^{\text{p}}] - \\
& \quad \frac{2}{15} \sum_{\mathbf{p}} g_2^4 Y_e^{i1i2} \text{LF}_{5,-2}[\mathbf{m}_q^{\text{p}}] - \frac{1}{18} g_2^4 Y_e^{i1i2} \text{LF}_{3,0}[\tilde{\mu}] - \frac{1}{12} g_2^4 Y_e^{i1i2} \text{LF}_{4,-1}[\tilde{\mu}] + \\
& \quad \frac{4}{45} g_2^4 Y_e^{i1i2} \text{LF}_{5,-2}[\tilde{\mu}] + \frac{1}{16} g_1^2 g_2^2 Y_e^{i1i2} \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_1^{\text{i}1}] - \frac{1}{16} g_2^4 Y_e^{i1i2} \text{LF}_{2,2,-1}[\mathbf{m}_2, \mathbf{m}_1^{\text{i}1}] + \\
& \quad \frac{2}{3} g_2^4 Y_e^{i1i2} \text{LF}_{2,1,0}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{3} g_2^4 Y_e^{i1i2} \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 Y_e^{i1i2} \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] - \\
& \quad \frac{1}{3} g_2^4 Y_e^{i1i2} \text{LF}_{3,1,-1}[\mathbf{m}_2, \tilde{\mu}] + \frac{5}{3} g_2^4 Y_e^{i1i2} \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] - \frac{3}{2} g_2^4 m_2^2 Y_e^{i1i2} \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] - \\
& \quad g_2^4 Y_e^{i1i2} \text{LF}_{3,3,-3}[\mathbf{m}_2, \tilde{\mu}] + g_2^4 m_2^2 Y_e^{i1i2} \text{LF}_{3,3,-2}[\mathbf{m}_2, \tilde{\mu}] - g_2^4 Y_e^{i1i2} \text{LF}_{4,2,-3}[\mathbf{m}_2, \tilde{\mu}] + \\
& \quad g_2^4 m_2^2 Y_e^{i1i2} \text{LF}_{4,2,-2}[\mathbf{m}_2, \tilde{\mu}] + g_2^2 Y_e^{i1i2} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_1^{\text{p}}, \mathbf{m}_e^{\text{r}}] - \\
& \quad \frac{1}{4} g_2^2 Y_e^{i1i2} \left(9 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{Y}_e^{\text{pr}} Y_e^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_1^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \\
& \quad \frac{1}{6} g_2^2 Y_e^{i1i2} \left(7 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{Y}_e^{\text{pr}} Y_e^{\text{pr}} \right) \text{LF}_{5,1,-2}[\mathbf{m}_1^{\text{p}}, \mathbf{m}_e^{\text{r}}] - \frac{1}{8} g_1^2 g_2^2 Y_e^{i1i2} \text{LF}_{2,1,0}[\mathbf{m}_1^{\text{i}1}, \mathbf{m}_1] + \\
& \quad \frac{1}{8} g_2^4 Y_e^{i1i2} \text{LF}_{2,1,0}[\mathbf{m}_1^{\text{i}1}, \mathbf{m}_2] + 3 g_2^2 Y_e^{i1i2} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] - \\
& \quad \frac{3}{4} g_2^2 Y_e^{i1i2} \left(9 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{Y}_d^{\text{pr}} Y_d^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \\
& \quad \frac{1}{2} g_2^2 Y_e^{i1i2} \left(7 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{Y}_d^{\text{pr}} Y_d^{\text{pr}} \right) \text{LF}_{5,1,-2}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] - \\
& \quad \frac{3}{4} \overline{Y}_d^{\text{pr}} Y_d^{\text{sr}} Y_e^{i1i2} \overline{Y}_u^{\text{st}} Y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}] + \frac{3}{4} \overline{Y}_d^{\text{pr}} Y_d^{\text{sr}} Y_e^{i1i2} \overline{Y}_u^{\text{st}} Y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}] - \\
& \quad \frac{1}{4} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} + 4 \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{2,2,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \quad \frac{3}{4} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \\
& \quad \frac{1}{4} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} - 2 \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \quad \frac{3}{4} g_2^2 Y_e^{i1i2} \left(-\overline{a}_u^{\text{pr}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] - \\
& \quad \frac{3}{4} \overline{Y}_d^{\text{pr}} Y_d^{\text{sr}} Y_e^{i1i2} \overline{Y}_u^{\text{st}} Y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{p}}] + \frac{3}{4} \overline{Y}_d^{\text{pr}} Y_d^{\text{sr}} Y_e^{i1i2} \overline{Y}_u^{\text{st}} Y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{p}}] + \\
& \quad \frac{1}{4} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} + 7 \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{3,1,0}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \quad \frac{1}{4} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} + 4 \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{3,2,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \quad \frac{3}{4} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} + 7 \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{4,1,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \quad \frac{1}{2} g_2^2 Y_e^{i1i2} \left(\overline{a}_u^{\text{pr}} a_u^{\text{pr}} + 7 \tilde{\mu}^2 \overline{Y}_u^{\text{pr}} Y_u^{\text{pr}} \right) \text{LF}_{5,1,-2}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \quad \frac{3}{2} \overline{Y}_d^{\text{pr}} Y_d^{\text{sr}} Y_e^{i1i2} \overline{Y}_u^{\text{st}} Y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] + \frac{3}{2} \overline{Y}_d^{\text{pr}} Y_d^{\text{sr}} Y_e^{i1i2} \overline{Y}_u^{\text{st}} Y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] - \\
& \quad g_1^2 g_2^2 Y_e^{i1i2} \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{2} g_1^4 \tilde{\mu}^2 Y_e^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] + \\
& \quad \frac{29}{12} g_1^2 g_2^2 Y_e^{i1i2} \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] + \frac{1}{2} g_1^4 \tilde{\mu}^2 Y_e^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] - \\
& \quad \frac{4}{3} g_1^2 g_2^2 Y_e^{i1i2} \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] - \frac{13}{3} g_2^4 Y_e^{i1i2} \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] + \frac{4}{3} g_2^4 Y_e^{i1i2} \text{LF$$